

VIRTUALWORKS LAB INTERNSHIP

PHARMACOVIGILANCE

TASK 3

CAUSALITY ASSESSMENT

Evaluation of Drug-ADR Relationship Using Standard Causality Methods

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Note: All patient cases in this report are fictional and are prepared only for internship learning and causality-assessment practice.

1. Aim

To evaluate whether an adverse event is related to a suspected medicine by reviewing the time relationship, known pharmacology, response after stopping the drug, rechallenge information and possible alternative causes.

2. Introduction

Causality assessment is an important part of pharmacovigilance. An adverse event occurring after a medicine does not automatically mean that the medicine caused it. The assessor has to look at the complete case and decide how strong the relationship is between the drug and the reaction.

For this task, I used the WHO-UMC causality approach as the main method because it directly uses categories such as Certain, Probable/Likely and Possible. I also reviewed the basic idea of the Naranjo algorithm as another structured method used to support ADR causality assessment.

3. Main Points Checked During Assessment

Time relationship	Did the reaction begin after the suspected medicine was started, or after a dose change?
Known reaction	Is the event consistent with the known pharmacological or safety profile of the medicine?
Dechallenge	Did the reaction improve after withdrawal or dose reduction?
Rechallenge	Did the reaction return after the medicine was given again, when such information was available?
Alternative causes	Could the disease, another medicine, food, infection or another condition explain the event?
Objective evidence	Are there laboratory findings or clinical observations that support the suspected reaction?

4. WHO-UMC Categories Used in This Task

Category	Practical meaning used for these cases
Certain	Strong time relationship; no reasonable alternative cause; reaction is pharmacologically/clinically convincing; dechallenge is compatible and rechallenge may support the relationship when necessary.
Probable / Likely	Reasonable time relationship; unlikely to be explained by another cause; improvement after withdrawal is clinically reasonable; rechallenge is not required.
Possible	Reasonable time relationship is present, but another disease, medicine or factor could also explain the reaction, or dechallenge information is unclear.

Note: The Naranjo scale uses Definite, Probable, Possible and Doubtful categories. Since the VirtualWorks task specifically asks for Certain, Probable and Possible, the final case labels below follow WHO-UMC terminology.

5. Case Study 1 - Certain

A 27-year-old male with bronchial asthma used salbutamol 200 micrograms by inhalation for wheezing. Within about 10 minutes he developed fine tremor of both hands and mild palpitations. The symptoms reduced within one to two hours. On two later occasions, when salbutamol was again required for asthma symptoms, the same tremor appeared soon after the dose and again resolved without treatment. He was not taking another medicine known to produce tremor and had no fever or hypoglycaemia.

Suspected drug	Salbutamol inhaler
Adverse reaction	Fine hand tremor with mild palpitations
Time relationship	Reaction appeared within minutes after each dose
Dechallenge	Symptoms resolved as the drug effect wore off
Rechallenge	Positive - the same reaction recurred after later medically required doses
Alternative cause	No convincing alternative cause identified
Final classification	Certain

My assessment: The reaction had a clear and reproducible relationship with salbutamol. Tremor and palpitations are pharmacologically consistent with beta-2 agonist activity, and the event returned after re-exposure. This supports a Certain relationship.

6. Case Study 2 - Certain

A 56-year-old female with type 2 diabetes received her usual dose of rapid-acting insulin before breakfast. Because of an administration error, she received a higher dose than intended. About one hour later she developed sweating, tremor, hunger and confusion. Capillary blood glucose was 46 mg/dL. She improved promptly after oral glucose and intravenous dextrose. Review of the case did not identify sepsis, alcohol use, renal failure or another glucose-lowering drug change.

Suspected drug	Rapid-acting insulin
Adverse reaction	Symptomatic hypoglycaemia
Objective evidence	Capillary glucose 46 mg/dL
Time relationship	Symptoms occurred about 1 hour after the excessive dose
Dechallenge / management	Recovered after glucose treatment and dose correction
Alternative cause	No better explanation identified
Final classification	Certain

My assessment: The reaction is pharmacologically expected after excessive insulin exposure and was confirmed by low blood glucose. The timing and objective evidence make the drug-reaction relationship sufficiently convincing for a Certain classification.

7. Case Study 3 - Probable / Likely

A 34-year-old male was started on amoxicillin + clavulanic acid 625 mg three times daily for acute bacterial sinusitis. Two days later he developed a red maculopapular rash over the chest and arms with itching. There was no breathing difficulty or mucosal involvement. The antibiotic was stopped and cetirizine was given. The rash improved over the next two days. He had no new food, cosmetic or other medicine exposure, and the suspected antibiotic was not restarted.

Suspected drug	Amoxicillin + clavulanic acid
Adverse reaction	Maculopapular rash with pruritus
Time relationship	Reaction started about 2 days after therapy began
Dechallenge	Positive - rash improved after withdrawal
Rechallenge	Not done
Alternative cause	No strong alternative cause identified
Final classification	Probable / Likely

My assessment: The time relationship was appropriate, the reaction is known with penicillin antibiotics, and improvement followed withdrawal. Because no rechallenge was performed, I classified the case as Probable/Likely rather than Certain.

8. Case Study 4 - Probable / Likely

A 49-year-old female with hypertension started enalapril 5 mg once daily. After about three weeks she developed a persistent dry cough, mainly at night. She had no fever, wheeze, sputum production or recent respiratory infection. Chest examination was normal. Enalapril was changed to an angiotensin receptor blocker, and the cough gradually disappeared over the following two weeks. Enalapril was not restarted.

Suspected drug	Enalapril
Adverse reaction	Persistent dry cough
Time relationship	Cough began about 3 weeks after starting treatment
Dechallenge	Positive - cough resolved after enalapril was stopped
Rechallenge	Not done
Alternative cause	Respiratory infection and other common causes not supported by case history
Final classification	Probable / Likely

My assessment: ACE-inhibitor cough is a well-recognized adverse effect. The compatible onset and improvement after withdrawal, together with the absence of a better explanation, support a Probable/Likely relationship.

9. Case Study 5 - Possible

A 45-year-old male with knee pain took diclofenac 50 mg twice daily for five days. During treatment he developed upper abdominal burning and dyspepsia. The symptoms improved after diclofenac was stopped and an antacid was used. However, the patient also reported a previous history of gastritis and frequent spicy meals during the same week. No endoscopy or other investigation was available.

Suspected drug	Diclofenac
Adverse reaction	Dyspepsia / epigastric burning
Time relationship	Symptoms appeared during diclofenac therapy
Dechallenge	Symptoms improved after stopping the drug, but antacid was also used
Alternative cause	Previous gastritis and diet could also explain the symptoms
Objective evidence	No specific investigation available
Final classification	Possible

My assessment: Diclofenac can cause gastric irritation, so the relationship is reasonable. However, the patient already had gastritis and dietary factors that could produce the same symptoms. Therefore, the event is best classified as Possible.

10. Case Study 6 - Possible

A 60-year-old female started atorvastatin 20 mg once daily for dyslipidaemia. About four weeks later she reported generalized muscle aching and tiredness. Creatine kinase was mildly elevated. At the same visit, thyroid testing showed untreated hypothyroidism. Atorvastatin was temporarily withheld while thyroid treatment was started. Her muscle symptoms improved, but because two changes were made at the same time, the exact contribution of atorvastatin could not be established.

Suspected drug	Atorvastatin
Adverse reaction	Myalgia with mild CK elevation
Time relationship	Symptoms appeared about 4 weeks after starting treatment
Dechallenge	Improvement occurred after statin withdrawal and thyroid treatment
Alternative cause	Untreated hypothyroidism is a plausible cause of muscle symptoms
Rechallenge	Not available
Final classification	Possible

My assessment: The timing is compatible with statin-associated muscle symptoms, but untreated hypothyroidism is a strong competing explanation. Because the improvement followed both statin withdrawal and thyroid treatment, a Possible classification is more appropriate.

11. Summary of Causality Assessments

Case	Suspected drug	Reaction	Key evidence	WHO-UMC classification
1	Salbutamol	Tremor / palpitations	Reproducible after repeated doses; no alternative cause	Certain
2	Rapid-acting insulin	Hypoglycaemia	Low glucose confirmed after excessive dose; prompt recovery	Certain
3	Amoxicillin + clavulanate	Maculopapular rash	Good timing and positive dechallenge; no rechallenge	Probable / Likely
4	Enalapril	Dry cough	Known reaction; resolved after withdrawal	Probable / Likely
5	Diclofenac	Dyspepsia	Compatible timing but gastritis/diet are alternatives	Possible
6	Atorvastatin	Myalgia	Compatible timing but hypothyroidism is an alternative	Possible

12. What I Learned From the Cases

The case studies show that causality is not decided only by whether a reaction happened after a medicine was taken. A Certain assessment needs very strong evidence, such as a pharmacologically convincing event or reproducible reaction. A Probable/Likely assessment can be made without rechallenge when the timing, known drug effect and dechallenge are convincing and other causes are unlikely. A Possible assessment is appropriate when the drug could be responsible but another explanation remains reasonable.

13. Conclusion

Causality assessment helps pharmacovigilance professionals judge the strength of the relationship between a suspected medicine and an adverse reaction. Using the WHO-UMC approach, the six fictional cases in this task were classified as two Certain, two Probable/Likely and two Possible reactions. Reviewing timing, dechallenge, rechallenge, known adverse effects and alternative causes made the classification more systematic and clinically meaningful.

14. References

1. World Health Organization - Uppsala Monitoring Centre. The use of the WHO-UMC system for standardized case causality assessment.
2. Naranjo CA, Busto U, Sellers EM, et al. A method for estimating the probability of adverse drug reactions. *Clinical Pharmacology & Therapeutics*. 1981;30(2):239-245.
3. Pharmacovigilance Programme of India (PvPI), Indian Pharmacopoeia Commission. Guidance and Suspected Adverse Drug Reaction Reporting Form.
4. Standard prescribing information / drug-safety references for salbutamol, insulin, amoxicillin-clavulanate, enalapril, diclofenac and atorvastatin.

Academic integrity note: All patient scenarios are fictional and have been created only for internship learning. No real patient identifiers or confidential medical information are included.